

Statistical and Machine Learning

Clustering analysis – Flipped Classroom Handout (Teacher Version)

Instructions

All questions should first be discussed in pairs or small groups. Afterwards, we shall discuss them together. Solutions will be distributed after each session.

Question 1: Clustering fundamentals

- (a) In your own words, what is the difference between **supervised** and **unsupervised** learning? Where does clustering fit?

Answer: In supervised learning, each observation comes with a label or response variable that the model is trained to predict. In unsupervised learning there are no labels; the goal is to discover structure in the data. Clustering is an unsupervised method: it finds groups of similar observations without using any pre-specified class labels.

- (b) A dataset has $N = 100$ observations in $\mathbb{R}^5$. You are given an assignment function $a(\cdot)$ that maps each observation to one of $K = 3$ clusters. State the two conditions that a good assignment function should satisfy in terms of within-cluster and between-cluster distances.

Answer: (i) Observations within the same cluster should have small pairwise distances: $a(x_i) = a(x_j) \Rightarrow d(x_i, x_j)$ small. (ii) Observations in different clusters should have large pairwise distances: $a(x_i) \neq a(x_j) \Rightarrow d(x_i, x_j)$ large. Equivalently, the assignment function satisfies $a(x_i) = a(x_j) \Leftrightarrow x_i$ and x_j are similar.

- (c) Give **two** examples of dissimilarity measures D_{ij} that can be used for quantitative variables.

Answer: Any two of: Euclidean distance $\|x_i - x_j\|_2$; Manhattan (city block) distance $\sum_k |x_{ik} - x_{jk}|$; Minkowski distance; Mahalanobis distance.

Question 2: Purity and Rand Index

Consider four clusters with $N = 20$ objects and true class labels A, B, C:

Cluster	Contents	N_i
1	A A A A B	5
2	B B B A	4
3	C C C C C B	6
4	A A C C C	5

- (a) Compute the purity p_i of each cluster, and the overall purity of the clustering.

Answer:

- Cluster 1: dominant class A with 4 out of 5, so $p_1 = 4/5$.
- Cluster 2: dominant class B with 3 out of 4, so $p_2 = 3/4$.
- Cluster 3: dominant class C with 5 out of 6, so $p_3 = 5/6$.

- Cluster 4: tie between A (2) and C (3), dominant class C with 3 out of 5, so $p_4 = 3/5$.

Overall purity:

$$\text{purity} = \frac{5}{20} \cdot \frac{4}{5} + \frac{4}{20} \cdot \frac{3}{4} + \frac{6}{20} \cdot \frac{5}{6} + \frac{5}{20} \cdot \frac{3}{5} = \frac{4 + 3 + 5 + 3}{20} = \frac{15}{20} = 0.75$$

- (b) Explain in one sentence the key **limitation** of purity as an evaluation metric.

Answer: Purity is trivially 1 if every observation is placed in its own singleton cluster, so it does not penalise having too many clusters.

- (c) For the Rand index, define TP , TN , FP , and FN in terms of pairs of objects. What familiar classification metric does the Rand index correspond to?

Answer: For every pair of objects: TP = pair is in the same cluster in both the estimated clustering U and the reference V ; TN = pair is in different clusters in both U and V ; FP = pair is in the same cluster in U but different clusters in V ; FN = pair is in different clusters in U but the same cluster in V . The Rand index $R = (TP + TN)/(TP + FP + FN + TN)$ is exactly the **accuracy** of pairwise clustering decisions.

- (d) Suppose a new clustering has $TP = 30$, $FP = 10$, $FN = 15$, $TN = 45$. Compute the Rand index R .

Answer:

$$R = \frac{TP + TN}{TP + FP + FN + TN} = \frac{30 + 45}{30 + 10 + 15 + 45} = \frac{75}{100} = 0.75$$

Question 3: Hierarchical Agglomerative Clustering

- (a) Describe in plain language what the **dendrogram** produced by HAC represents. What does the height of a node correspond to?

Answer: A dendrogram is a binary tree that records the entire history of merges. Each leaf corresponds to a single observation, and each internal node represents a merge event. The height of an internal node is the dissimilarity at which the two sub-trees were merged. Cutting the dendrogram horizontally at height h reveals all clusters formed before dissimilarity h .

- (b) You are given five points and told that single-link clustering is used with city block distance. The first merge occurs between points (2, 4). Explain what this means about the distances in the data.

Answer: The pair (2, 4) has the smallest city block distance among all $\binom{5}{2} = 10$ pairs. Under single linkage, the inter-cluster distance equals the minimum pairwise distance between members, so these two points are closer to each other (in city block distance) than any other pair of points.

- (c) Complete the table below comparing the three linkage criteria:

Linkage	Formula for $d(G, H)$	Properties*	Scale invariant?
Single	$\min_{i \in G, i' \in H} d_{i, i'}$	Elongated, chained clusters	Yes
Complete	$\max_{i \in G, i' \in H} d_{i, i'}$	Compact clusters; sensitive to outliers	Yes
Average	$\frac{1}{n_G n_H} \sum_{i \in G} \sum_{i' \in H} d_{i, i'}$	Compact and well-separated	No

- By Properties, we refer to the shape of clusters (i.e., elongated, compact, well-separated, etc) as well as to the properties of the method (i.e., sensitive to outliers)

- (d) Give **one advantage** and **one disadvantage** of HAC compared to K-means.

Answer: Advantage: HAC does not require specifying K in advance; the dendrogram allows exploring any number of clusters. Disadvantage: HAC merges are irreversible (a wrong early merge cannot be undone), and it is computationally expensive at $O(N^2)$ – $O(N^3)$, scaling poorly to large datasets.

Question 4: K-means clustering

- (a) Write down the **distortion** objective $J(\mathbf{M}, \mathbf{Z})$ minimised by K-means. What are $\mathbf{M}$ and $\mathbf{Z}$?

Answer:

$$J(\mathbf{M}, \mathbf{Z}) = \sum_{n=1}^N \|\mathbf{x}_n - \boldsymbol{\mu}_{z_n}\|^2 = \|\mathbf{X} - \mathbf{Z}\mathbf{M}^\top\|_F^2$$

$\mathbf{M} \in \mathbb{R}^{D \times K}$ has the K cluster centroids $\boldsymbol{\mu}_k$ as its columns. $\mathbf{Z} \in [0, 1]^{N \times K}$ is the assignment matrix, where $z_{nk} = 1$ if point n belongs to cluster k and 0 otherwise.

- (b) Describe the two steps of the K-means algorithm (assignment step and update step). Why is the algorithm guaranteed to converge?

Answer: Assignment step: assign each point x_n to the nearest centroid, $z_n^* = \arg \min_k \|x_n - \mu_k\|_2^2$. Update step: recompute each centroid as the mean of its assigned points, $\mu_k = \frac{1}{N_k} \sum_{n: z_n=k} x_n$. Convergence is guaranteed because each step can only decrease or leave unchanged the distortion J , and J is bounded below by 0, so the sequence of J values must converge. Since the number of possible assignments is finite, the algorithm terminates.

- (c) Why does the distortion J **not** provide a valid criterion for choosing the number of clusters K ? What does the **elbow method** suggest instead, and why is it sometimes unreliable?

Answer: J decreases monotonically with K (trivially reaching zero at $K = N$), so minimising J always favours more clusters. The elbow method plots J vs K and looks for a “kink” or “elbow” where the rate of decrease slows sharply; K at the kink is selected. It is unreliable in practice because the kink is often subtle or absent, making it hard to identify objectively.

- (d) The **silhouette coefficient** for point i is $sc(i) = \frac{b_i - a_i}{\max(a_i, b_i)}$. Interpret the three boundary cases $sc(i) \approx +1$, $sc(i) \approx 0$, and $sc(i) \approx -1$.

Answer: $sc(i) \approx +1$: point i is close to members of its own cluster and far from other clusters, so it is well-clustered. $sc(i) \approx 0$: point i is near the boundary between two clusters ($a_i \approx b_i$). $sc(i) \approx -1$: point i is closer on average to another cluster than to its own, suggesting it may be misassigned.

Question 5: Mixture Models and the EM Algorithm

- (a) Write down the density of a **Gaussian mixture model (GMM)** with K components. Identify all parameters and state the constraint on the mixing weights π_k .

Answer:

$$p(\mathbf{x} \mid \boldsymbol{\theta}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

Parameters: mixing weights $\boldsymbol{\pi} = (\pi_1, \dots, \pi_K)$, means $\boldsymbol{\mu}_k \in \mathbb{R}^D$, and covariance matrices $\boldsymbol{\Sigma}_k \in \mathbb{R}^{D \times D}$ (positive semi-definite). Constraint: $0 \leq \pi_k \leq 1$ for all k and $\sum_{k=1}^K \pi_k = 1$.

- (b) Define the **responsibility** r_{nk} in a GMM. What does it represent, and why is it called a “soft” assignment?

Answer:

$$r_{nk} \triangleq p(z_n = k \mid \mathbf{x}_n, \boldsymbol{\theta}) = \frac{\pi_k \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{k'=1}^K \pi_{k'} \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_{k'}, \boldsymbol{\Sigma}_{k'})}$$

It is the posterior probability that observation n was generated by component k . It is called a “soft” assignment because each point is fractionally assigned to all K clusters simultaneously (with weights summing to 1), rather than being assigned entirely to one cluster as in K-means.

- (c) Briefly describe what happens in the **E step** and the **M step** of the EM algorithm when fitting a GMM. What quantity is guaranteed to be non-decreasing at each iteration?

Answer: E step: given current parameters $\theta^{(t)}$, compute the responsibilities $r_{nk}^{(t)}$ for every point and cluster using Bayes’ rule. M step: update the parameters by maximising the expected complete-data log likelihood weighted by the responsibilities; this yields closed-form updates for μ_k , Σ_k , and π_k . The observed-data log likelihood $\ell(\theta)$ is guaranteed to be non-decreasing at each iteration (monotonic convergence).

- (d) State the **two approximations** that reduce EM for a GMM to the K-means algorithm.

Answer: (i) Fix $\Sigma_k = \mathbf{I}$ and $\pi_k = 1/K$ for all k (spherical, equal-weight clusters, so only the means μ_k need to be estimated). (ii) Replace the soft responsibilities with hard assignments in the E step: $r_{nk} \approx \mathbb{I}(k = z_n^*)$ where $z_n^* = \arg \max_k r_{nk}$. With both approximations the M step reduces exactly to the K-means centroid update.

Question 6: Spectral Clustering

- (a) Why do K-means and GMMs struggle with **non-convex** clusters? Give an example of a dataset shape where spectral clustering would be expected to outperform K-means.

Answer: K-means minimises Euclidean distance to centroids, implicitly assuming spherical clusters; GMMs assume each cluster is generated by a Gaussian, which also produces elliptical (convex) regions. Both therefore fail when clusters have non-convex shapes. Example: two interleaved rings (concentric circles) or two interleaved spirals, where points in the same ring are far apart in Euclidean space but connected through the ring structure.

- (b) Define the **graph Laplacian $\mathbf{L}$** in terms of the weight matrix $\mathbf{W}$ and the degree matrix $\mathbf{D}$. What is the degree d_i of node i ?

Answer: $\mathbf{L} = \mathbf{D} - \mathbf{W}$, where $\mathbf{W}$ is the symmetric similarity (weight) matrix and $\mathbf{D} = \text{diag}(d_1, \dots, d_N)$ is the diagonal degree matrix. The degree of node i is the total weight of all edges connected to it: $d_i = \sum_{j=1}^N w_{ij}$.

- (c) State Theorem 21.5.1 about the eigenspace of $\mathbf{L}$ with eigenvalue 0. What does this theorem tell us about the relationship between eigenvectors and cluster structure?

Answer: Theorem 21.5.1: the eigenspace of $\mathbf{L}$ with eigenvalue 0 is spanned by the indicator vectors $\mathbf{1}_{S_1}, \dots, \mathbf{1}_{S_K}$, where $S_1, \dots, S_K$ are the K connected components of the graph. This means that when the graph has K perfectly separated components, the K zero-eigenvectors of $\mathbf{L}$ encode exactly which nodes belong to which cluster. The K eigenvectors with smallest eigenvalues therefore act as low-dimensional feature vectors that capture cluster membership, which is why clustering in this eigenvector space (via K-means) recovers the graph’s connected components.

- (d) List the steps of the **Ng–Jordan–Weiss** spectral clustering algorithm.

Answer:

1. Build the similarity matrix $\mathbf{W}$ (e.g. Gaussian kernel $w_{ij} = \exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / 2\sigma^2)$).
2. Compute the symmetric normalised Laplacian $\mathbf{L}_{sym} = \mathbf{D}^{-1/2} \mathbf{L} \mathbf{D}^{-1/2}$ and find its K eigenvectors with smallest eigenvalues; stack them as columns of $\mathbf{U} \in \mathbb{R}^{N \times K}$.
3. Row-normalise $\mathbf{U}$: set $t_{ij} = u_{ij} / \sqrt{\sum_k u_{ik}^2}$, forming matrix $\mathbf{T}$.
4. Apply K-means to the rows $\mathbf{t}_i \in \mathbb{R}^K$ of $\mathbf{T}$.
5. Map the cluster assignments back to the original data points.

Question 7: Comparing clustering methods

Property	HAC	K-means	GMM + EM	Spectral
Requires K in advance	No	Yes	Yes	Yes
Has a well-defined objective	No	Yes (J)	Yes (ℓ)	Yes (Ncut)
Produces soft (probabilistic) assignments	No	No	Yes	No
Handles non-spherical clusters	Partly	No	Yes (full Σ_k)	Yes
Deterministic (no random initialisation)	Yes	No	No	Partly
Scales well to large N	No ($O(N^3)$)	Yes ($O(NKT)$)	Moderate	No ($O(N^3)$)

Justification for “Handles non-spherical clusters”: K-means assigns points to their nearest centroid using Euclidean distance, which creates spherical Voronoi regions; it cannot adapt to elongated or curved cluster shapes. GMM with EM uses full covariance matrices Σ_k that can be elliptical, so it handles non-spherical clusters as long as they are roughly Gaussian. Spectral clustering works in the eigenvector space of the graph Laplacian, which encodes connectivity rather than Euclidean distance, allowing it to correctly separate non-convex clusters such as rings or spirals that no centroid- or Gaussian-based method can recover.